



Arctic Arthropods: A Review of Systematics and Ecology with Particular Reference to the North American Fauna, Soil and Permafrost Surveys in the Arctic, the Last Great Ice Sheets, the Glaciation of the Ecuadorian Andes

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BOOK REVIEWS

ARCTIC ARTHROPODS: A REVIEW OF SYSTEMATICS AND ECOLOGY WITH PARTICULAR REFERENCE TO THE NORTH AMERICAN FAUNA. By H. V. Danks. Ottawa: Entomological Society of Canada, 1981. viii + 608 pp. Available from Entomological Society of Canada, 1320 Carling Avenue, Ottawa, Ontario K1Z 7K7. \$35.00 Can (Society members \$24.50) plus \$2.00 postage and handling.

This is an encyclopedic account of the state of knowledge of arctic arthropods, excluding the fully aquatic Crustacea. As indicated by the title, the thrust is on the North American fauna, yet generalizations apply to arctic arthropod biology. This is especially true for Part I, an overview of arctic environments. Herein arctic regions are described from their physiographic and glacial histories. Climates, microclimates, permafrost, and soils are briefly described with good references to detailed studies. The biological features are also succinctly discussed, leading to general conclusions which summarize the factors influencing arctic life, its diversity and zonation. The level of these useful generalities makes them simple, so readers with specific questions should consult primary sources. Danks provides some 500 citations and Appendix 1 for a synopsis of arctic islands.

Part II, "The Arthropod Fauna and its Relationships," is the meat, representing about one-half the volume. Danks introduces the subject through a history of arthropod collection and proceeds to caveats in interpretation as results of taxonomic problems, paucity of specimens, insufficient information, and inadequate range records. Species variation is covered separately; both variable and nonvariable species are discussed. Danks has assembled a prodigious amount of available information to give a detailed, comprehensive, and accurate picture as possible of the arctic fauna, its composition (2104 species), and patterns of distribution. His breakdown is to family, each described in terms of their lives and distributions. A thorough account of distribution patterns is given, and illustrated by 62 maps. Nevertheless, Danks concludes that faunal range limits tend to appear at the northern limit of closed boreal forest, and longitudinally at Hudson Bay and, less evidently, at the MacKenzie River. Latitudinal zonation is not well defined, yet a southern arctic-northern arctic boundary is indicated. Greenland shows greater affinities with the nearctic than was previously supposed. No final map in the series, summarizing the distributional areas which are recognized and used extensively in Part III, is provided.

Within Part II, the section "Ecological Considerations" is especially valuable and integrates previous discussion. Terrestrial and freshwater habitats are examined together with the organisms and their adaptations; these are structural, behavioral, and metabolic. Winter survival and adaptive (often protracted) life-cycles are particularly

important and thoroughly discussed along with generalist tendencies in food ranges. Saprophagy seems widespread, and starts discussion of trophic relationships of arthropods and their places in interdependent ecosystem components. Herbivory (and flower relationships), ectoparasitism, parasitoides, predation, and disease are all well treated. The question of competition is raised and conflicting evidences from entomology and ornithology are given. Energy cycling is a difficult area in which to draw generalizations. Population biology of arctic arthropods remains elusive because of heterogeneity of habitats, year to year variation, and the need for long-term detailed studies.

There are 20 pages on relationships of arthropods and man on tundra, the bulk on biting flies. Despite the economic importance of these insects, much remains to be understood.

At the end of the ecological review, Danks returns to a biosystematic view in discussing the historical and ecological determinants of arthropod distribution. The impacts of refugia, postglacial changes, and arthropod dispersal are well synthesized. The conclusions are cautiously approached, with emphasis on the balance between historical and ecological constraints to arthropod dispersion which is needed to achieve explanation.

In conclusion, Danks stresses the continuing need for collecting over the entire arctic to increase understanding of ranges, variations, and diversity. Arctic arthropods are not well understood, and clearly provide a fruitful ground for future research. Needs are for organized and long-term research, especially at selected sites, as well as in the arctic in general, using standardized and thorough techniques.

Part III of this work is a 144-page systematic list of reported North American arctic arthropods and their distributions. The references for both Parts II and III are combined and contain about 2000 citations.

Danks has accomplished a remarkable task. With competence and balanced judgement, he has authoritatively synthesized an extensive and diffuse literature. He has arrived at some new conclusions, modified some old ones, and informatively pointed the way to future research. In short, this is a seminal work which will be a lasting contribution (well bound for durability) and which, although technically entomological and not light reading, should be part of every arctic biologist's reading and reference. I congratulate Dr. Danks and The Entomological Society of Canada for producing this scholarly work.

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SOIL AND PERMAFROST SURVEYS IN THE ARCTIC. By Kenneth A. Linell and John C. F. Tedrow. New York and Oxford: Clarendon Press, Oxford University Press, 1981. (Monographs on Soil Survey.) viii + 279 pp. \$49.50. ISBN 0-19-857557-2

A considerable amount of geotechnical and design information exists for permafrost areas in the Northern Hemisphere. In North America much of this information has been acquired in the last 15 to 20 years and especially during the last decade in response to accelerated resource exploration and development. Much of the published research, including design manuals and site selection criteria, have come from the Cold Regions Research Engineering Laboratories (CRREL and formerly SIPRE—Snow, Ice Permafrost Research Engineering Laboratory), as well as from the Division of Building Research (Canada). There have been, however, relatively few attempts to integrate site selection criteria including geotechnical considerations with the permafrost environment as a whole. Muller's early volume on permafrost (1947) is still a useful example. Within the last year two such books have appeared. The authors of the volume being reviewed here are both internationally recognized authorities on soils (in the broadest sense) and engineering problems in permafrost terrains. Their stated objective in *Soil and Permafrost Surveys in the Arctic* was "to focus on the special problems associated with surveying Arctic soils and permafrost directing attention to field problems and procedures."

The volume consists of 17 subject matter headings (chapters). The first six deal with the physical geography of permafrost climates, i.e., their extent, hydrologic and climatic characteristics, the composition of perennially and seasonally frozen ground, ground patterns, and soil (active layer) characteristics. Six of the next seven chapters deal with geotechnical and engineering characteristics of seasonally and perennially frozen soils, thermal stability of permafrost, engineering soil classification, and general and specific engineering problems. The remaining four chapters are concerned with the planning and execution of soil and permafrost surveys.

Engineering soil tests are listed in one of two very brief narrative appendices. Monitoring of soil and foundation behavior is treated in a few paragraphs in the second. A third brief appendix provides a list of some technical publications that give background on arctic soils and permafrost.

In fulfilling their stated objective the authors are only partially successful. For this reviewer the major problems lie in the organization of the subject headings and the unevenness of subject treatment. As examples, a section on northern agriculture and soil conservation appears between sections dealing with geotechnical considerations with little apparent relevance to either chapter. A section (chapter) titled Soil-Mechanics consists of only 2½ pages and is little more than an introduction to later sections. The thermal regime of perennially frozen ground is

treated in some considerable detail in one section while in a separate section only a light brush is applied to the thermal stability of permafrost.

The word soil in the title is somewhat misleading or at least subject to individual interpretation. It would have been better had the word engineering been included, for this is really the thrust of the volume.

The book has relatively few photographs, but is reasonably well provided with graphs, other line drawings, and tables. Because of the size of the book (22 × 14.5 cm) many of the tables and a few of the other illustrations have been reduced to the threshold of readability, and in a few cases beyond. This is especially irksome in view of the price.

On a more positive note the volume presents a general overview of some of the special problems in engineering site evaluation and placement in permafrost climates; however, there is little consideration of case histories with regard to field problems and procedures. On the whole the book will be useful to those new to permafrost regions and/or those in need of an introduction to the physical characteristics of perennially frozen soils. The price (\$49.50) will, however, discourage many, especially those younger scientists and engineers most in need of its contents.

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THE LAST GREAT ICE SHEETS. Edited by George H. Denton and Terence J. Hughes. New York: John Wiley, 1981. 484 pp + 27 maps slip-cased. \$95.00.

The Last Great Ice Sheets is an attempt to summarize the existing knowledge of the global distribution of ice sheets, ice caps, and valley glaciers, primarily during the late Wisconsin/Weichselian glacial stade (ca. 24 to 12 ka BP), and to use this information as input to glaciological models that might explain ice-sheet dynamics and past changes in the volume of water removed from the oceans and stored on land. The first three chapters are syntheses of published material on the extent of ice masses in Eurasia (B. Andersen), North America (P. Mayewski, G. Denton, T. Hughes), and local glaciers (J. Hollin, D. Schilling), followed by glaciological models for small (Schilling and Hollin) and large (Hughes) ice masses. For some reason, the ice extent in Antarctica is relegated to a later chapter, even though the numerical models are clearly based on observations from Antarctica. Chapter 6 provides an interpretive global overview for the last gla-

ciation and chapter 8 is an elaboration of earlier models for a late Wisconsin circum-Arctic ice sheet. Clearly, the editors have assumed an ambitious task; an attempt to summarize *and* evaluate the available literature of ice extent during the late Quaternary. There are noticeable differences in the treatment of different regions; hence the chapter on Europe contains no references more recent than 1976, whereas other chapters contain 1980 references. The chapter on Antarctica contains new data and summarizes many years of research by the authors, whereas the North American chapter is primarily the review of others' data.

One of the aims of this work was to depict the ice distribution on the earth's surface at the height of the last glacial maximum, 18 ka BP. Because of conflicts in published ice limits, the contributors chose to depict two models of the ice limits at this time, a so-called minimalist and maximalist version, although they make no pretense as to their preference for the maximum version. The glacial limits and theoretical ice-sheet reconstructions are presented as a series of maps both within the body of the text and on large loose sheets accompanying the book. Comparing the ice-terminal positions relative to the maximum and minimum reconstructions reveals that over very large stretches the differences are slight. The major discrepancies occur for the most part across Arctic Canada and Arctic Eurasia. Consequently, it is surprising that none of the researchers working in these areas is a contributor to the volume.

Substantial portions of chapters 1, 2, and 6 are devoted to regions of conflict in the field evidence. In their discussion of high-latitude ice limits, perhaps the most striking trend is the authors' constant attempt to critically evaluate in the greatest detail all evidence for limited ice, including reinterpretation of published results (e.g., pp. 103-105), yet to uncritically accept the evidence favoring the maximum ice limits. Some conflicts are readily apparent. For example, the lack of weathering of striae is accepted as evidence of a late Wisconsin age for the associated glacial event (e.g., pp. 130, 293), whereas the subdivision of till sheets on weathering characteristics of erratic boulders at sites from Newfoundland to the High Arctic is rejected. Yet the subdivision and correlation of Antarctic drift sheets (the author's main geological contribution) is based primarily on the weathering of erratics, moraine morphology, and the degree of pedogenesis (e.g., pp. 332, 338, 365-368). The reliability of Northern Hemisphere U-series dates on molluscs are seriously questioned (e.g., pp. 112, 113), but are accepted for Antarctica (pp. 364, 425). Throughout the discussion of North American ice margins, the reader is painfully aware that the authors believe the maximum reconstruction, with a marine-based ice sheet essentially in phase around its perimeter is most appropriate. In chapter 2 (p. 141) the authors state: "Most sectors of the North American ice sheets, with the possible exception of those in the Arctic, attained a late Wisconsin maximum between 21,000 and 14,000 BP." This claim is made despite the evidence

just summarized in the preceding 140 pages of the book that less than one-third of the continental ice sheet margins can be actually dated and have reached maxima within the time range.

The fundamental question raised here is whether all margins of the ice sheets responded synchronously, or whether there is a fundamental asynchrony between different geographic regions. The authors strongly favor the concept of the maximum ice sheet reconstruction with synchronous behavior around the ice sheet perimeters. The reason for this becomes apparent in the introduction to the glaciological modeling chapter.

Chapter 5 presents an in-depth appraisal of Hughes's numerical model for ice sheet reconstructions. In the introduction, he carefully presents the assumptions that must be fulfilled in order to validate the model. Of fundamental importance here are the prerequisites that (1) all margins of ice sheet must have attained maximum positions at the same time and (2) that at this time, the ice sheet approximated steady-state equilibrium conditions. If these assumptions are not met, then the model calculations are invalid.

Inherent in the minimalist ice reconstruction is the concept of fundamental asynchrony between the responses of the northern and southern Northern Hemisphere ice-sheet margins. Hence the underlying need in this book is to refute these claims and verify that all margins are time equivalent. The minimum model also suggests the ice sheets consisted of dynamic, semi-independent dispersal centers that may never have been in steady-state equilibrium.

The minimalists argue that climate, particularly precipitation, exerts a major control on high-latitude glacial activity and that decreased high-latitude precipitation during low-latitude glacial maxima would not allow major high-latitude glacier expansion at these times. This concept is lent support by the Antarctic data suggesting that the *climatically*-controlled alpine glaciers did not respond in phase with the proposed global ice volume record, and that synchronous behavior of the Antarctic Ice Sheet is explained by an eustatic regulating mechanism. The inference is that the high-latitude Southern Hemisphere climate was not conducive to glacial expansion 18 ka ago. Is there any reason to expect the northern high-latitude regions to have been any different?

The claim is made (p. 275) that arctic marine ice sheets are controlled by eustasy, and that Arctic glaciers expanded in response to lowered sea levels. However, the important variable is *relative* sea level, and in most glaciated Arctic regions, ice-contact marine deposits indicate isostatic depression of 75 to 150 m, similar to the maximum eustatic lowering and resulting in a net relative sea level change of only minor significance.

Despite what I perceive to be substantive flaws in logic and presentation, the volume offers major contributions as well. The concept of marine-based ice sheets, their inherent instability and interaction with ice shelves, although not new, is developed to a new level of under-

standing. My only regret is that the authors confined themselves within the time constraints of the "last glacial maximum." All agree that extensive glaciation has occurred in many north polar regions and that such an event must have been marine based. Conceptually, the glaciological arguments are not in contention. The force of these arguments would have been more universally compelling if divorced from temporal constraints.

Chapter 7 contains the most comprehensive and up-to-date overview of the glacial history of Antarctica. One cannot help feeling that the authors are on firmer ground when summarizing the information from their own field areas than when evaluating the regions for which they are literature dependent.

This volume stands as a testament to the still diverse opinions among glacial geologists and glaciologists on the magnitude, extent, and timing of even the most recent of the Pleistocene glacial stades. Even in mid-latitude regions that have been mapped and studied for over a century "state-line unconformities" are common and chronologies are in flux. Most high-latitude regions have known only two decades of intensive study, and the opinions diverge more dramatically. Because the glaciological models employed require a firm geological base as input, accurate reconstructions will require more field work in many areas. In the meantime, the output of the theoretical models can kindle the imagination of field workers and focus their attention on the types of evidence required to further refine the models.

My final impression is a rejoinder for all of us to not let the tail wag the dog; a theoretical model must be adjusted to fit the geological evidence, not the other way around.

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RELIKTOVYE STEPNYE KOMPLEKSY SEVERO-VOSTOCHNOI AZII [Relict steppe complexes of northeast Asia]. By B. A. Yurtsev. Novosibirsk: Nauka (Siberian Section), 1981. 168 pp. 2 rubles, 70 kopeck.

The ecology of the present flora and vegetation of the lands now bordering Beringia, that 1500-km-wide dry land connection between North America and northeastern Asia during the latest (and earlier) glaciation, is basic to our understanding the migrations of man, mammals, and other living things between these continents. Yurtsev's latest book follows closely on the recent Wenner-Gren Symposium in Austria (Stanley, 1980) on the Arctic steppe mammoth biome. He discusses and uses many of the papers that will be printed in the forthcoming proceedings volume edited by Hopkins, Mathews, Schweger, and Young. In his book he reviews the botanical evidence from the Soviet side.

The book consists of "Preface," "Introduction," "Steppe as a type of vegetation and its cryophyte analogues" (6 pp.), "Northeastern Yakutia" (25 pp.), "The Kolyma Basin" (forested part) (5 pp.), "Anadyr Basin" (5 pp.), "Chukotsk tundra" (23 pp.), "Structure and distribution of relict steppe complexes in the present and past" (24 pp.), "Discussion of questions on the paleogeography of Beringia" (13 pp.), References (8 pp, ¼ in English), Appendix (45 pp.), and Maps of species distributions (12 pp., 20 species or species groups, latitude and longitude net, scale 1:33 × 10⁶). The Appendix consists of 62 geobotanical descriptions (stand surveys, releves). As well as the vascular plants, mosses and lichens are included. Tabulations of 30 stands into species-stand matrices (association tables) make an overview of the data easier. Abundance is indicated by the Drude scale still used in the USSR, although extensively criticized, and by projected cover.

Landscape photographs in the Indigirka River basin could have been taken in Alaska-Yukon-northern British Columbia, except that larch is the zonally dominant conifer of Siberia.

Yurtsev suggests changes in the taxonomy and nomenclature of many species. An index to page references for the American steppe species is available from the reviewer.

Yurtsev classifies 142 of his Yakutian steppe species in relation to cold, moisture, snow cover, rockiness of habitat, and salinity. Well over half of these species occur in northwestern North America or have close relatives there. Many of the alpine and arctic-alpine species are calciphytes.

Climatic data for 19 stations invite comparisons with Alaska (and with the Wisconsin glaciation?). Yearly mean temperatures are -10 to -17°C, absolute maxima 30 to 35°, absolute minima -45 to -65°, precipitation 140 to 300 mm yr⁻¹ with 25 to 56 mm in July (July percent greater the lower the total).

Yurtsev emphasizes the ecological and therefore floristic differences between his tundra-steppe and polar desert communities (pp. 69ff.). Mesic plants characterize the latter. Steppe vegetation occurs as nonzonal islands in both the taiga and the tundra. This fact comes out strongly in the Appendix, which now provides a factual basis for discussions of the species composition and therefore ecological relationships in the widest sense. Although Yurtsev emphasizes the communality of the steppe vegetation of northeastern Asia with the North American Great Plains, such species common to both as *Koeleria "cristata," Carex duriuscula* (as *C. eleocharis* and *C. stenophylla* ssp. *spaniocarpa*), *C. obtusata*, *Ceratoides* spp., *Pulsatilla multifida*, *Chamaerhodos erecta*, *Oxytropis splendens*, *Astragalus aboriginum*, *A. tenellus*, *Artemisia frigida*, etc., are mountain, even alpine, plants in the American western mountains.

There has been, and continues to be, much discussion of the kind(s) of vegetation in Beringia when it was the land connection for migration between northeastern Asia

and North America. Yurtsev's book is a basic, important, indispensable, illuminating, excellent contribution to that discussion.

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THE GLACIATION OF THE ECUADORIAN ANDES. By Stefan Hastenrath. Rotterdam: A. A. Balkema, 1981. xiv + 159 pp., illustr., bibliogr. ISBN 90 6191 0382. Cloth Hfl. 50, \$24.00, £10, DM 48.

In reading S. Hastenrath's *Glaciation of the Ecuadorian Andes*, I unsuccessfully tried to imagine the readership for which this book was written. Correctly labeled as an inventory, this book lists the present, historical, and past extent of all ice fields and glaciers in the Eastern and Western Cordillera of Ecuador, based on extensive field observation, historical documents, and mapped moraine sequences, respectively. Abundant photographs and maps illustrate the individual description of each mountain and Appendices document the various sources from which the relevant information was

obtained. While the documentation of such basic data is always highly desirable for the specialist in the field, it remains of little use for the wider scientific community which is unable to judge the implications and meaning of the data without critical interpretation. It would have been desirable if the author had followed his own recommendations listed on the final text pages of the book and established a regional glacial chronology of Ecuador before suggesting a world-wide synopsis of past glacial fluctuations in the tropics. Along the same lines, the chapter on "Pleistocene Changes of Vegetation and Climate in Northern South America," an almost verbatim reproduction of van der Hammen's 1974 article in the *Journal of Biogeography*, does not add any relevant information to the book. This is, first, because no direct correlation is expected a priori to exist between the Ecuadorian glacier fluctuations and the vegetation changes in the Colombian intermontane basin, from where van der Hammen's data are obtained; and second because Hastenrath's glaciologic data do not yet provide the basic chronologic information necessary for such comparison. As an inventory the book is of high quality thanks to the critical presentation of the data and it is hoped that the future research of integration of the information and presentation of a synoptic picture will be equally well done.

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